

Pharyngeal Apparatus

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- The **pharyngeal apparatus** consists of:

- pharyngeal arches
- pharyngeal pouches
- pharyngeal clefts (grooves)
- pharyngeal membranes.

All these structures largely contribute to the formation of head & neck region (e.g., face, neck, definitive mouth, pharynx, and larynx).

- Early in the 4th week of intrauterine development, a series of surface elevations appear in the lateral wall of primitive pharynx caudal to stomatodeum.
- These elevations are termed **pharyngeal arches**
- The pouches between them on the inner aspect of the pharyngeal wall are termed **pharyngeal pouches**
- grooves between them on the outer aspect of the pharyngeal wall are called **pharyngeal clefts**

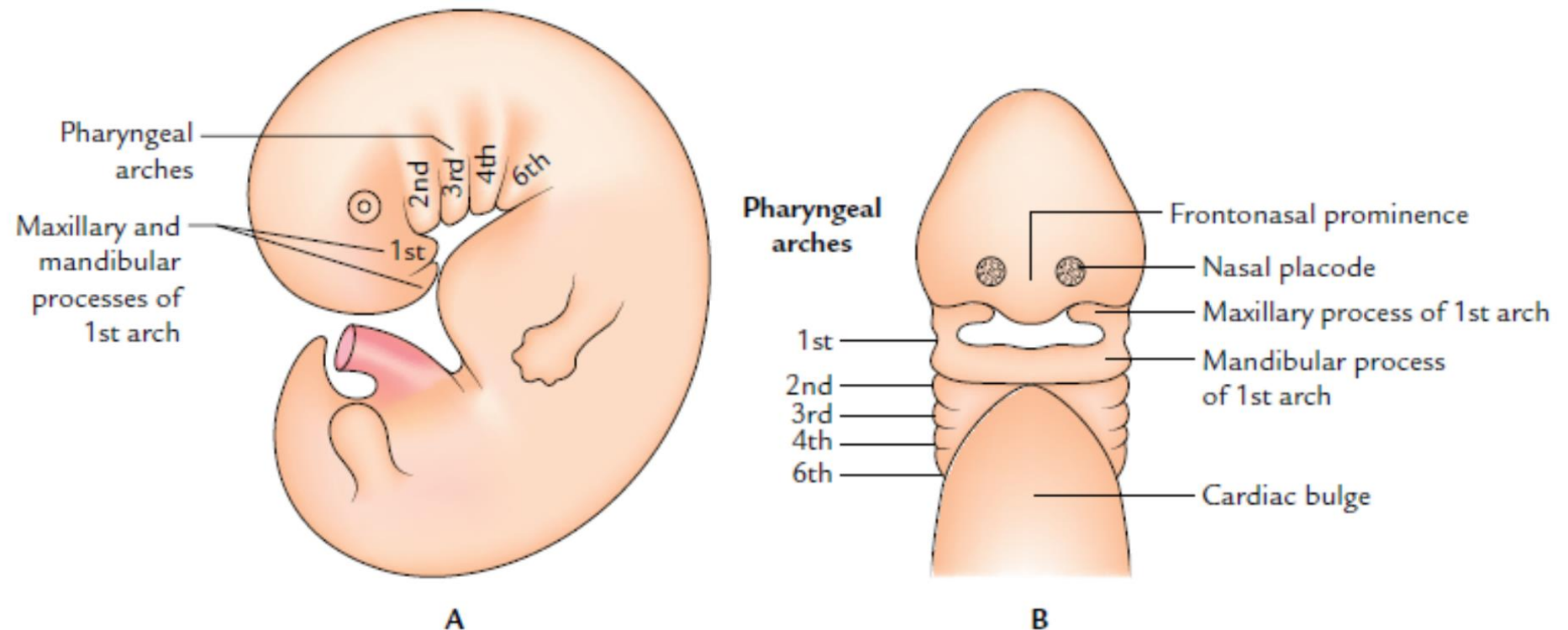
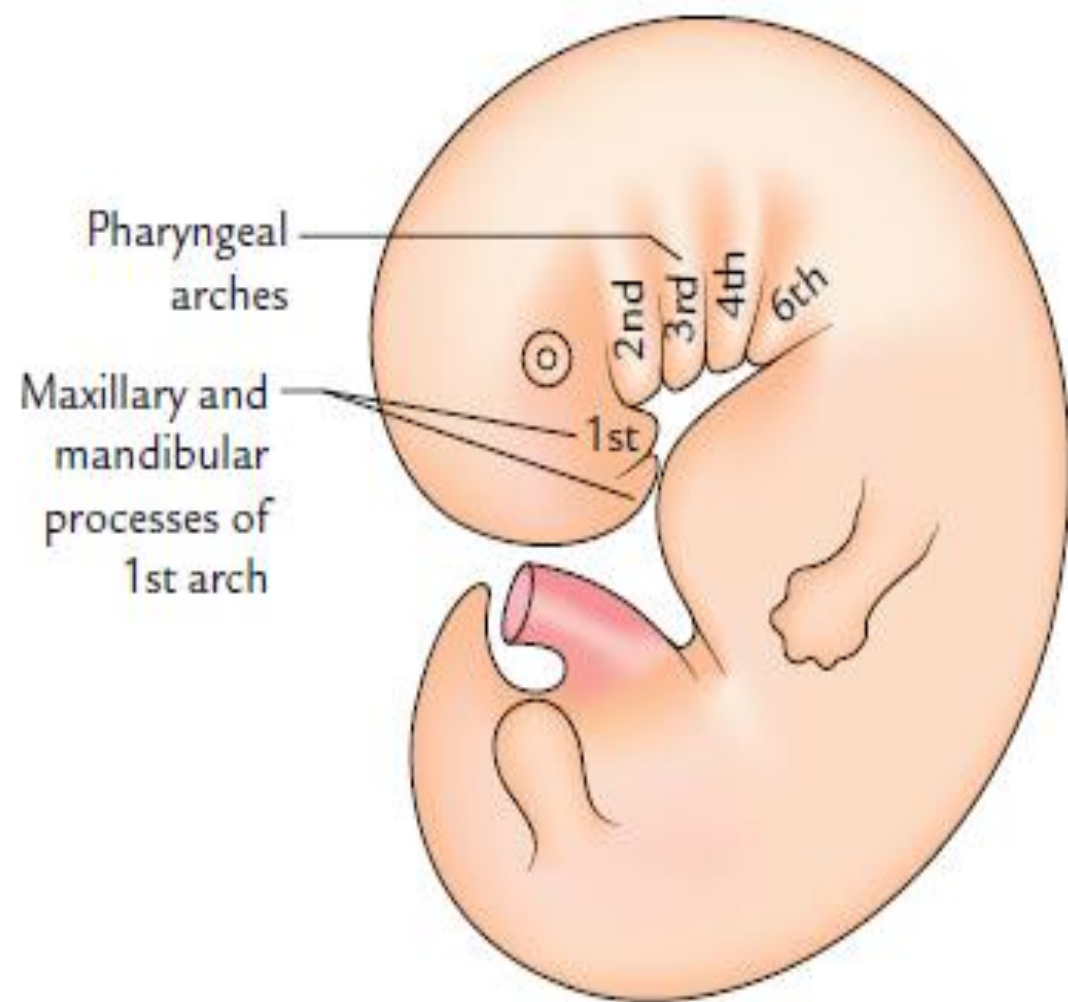
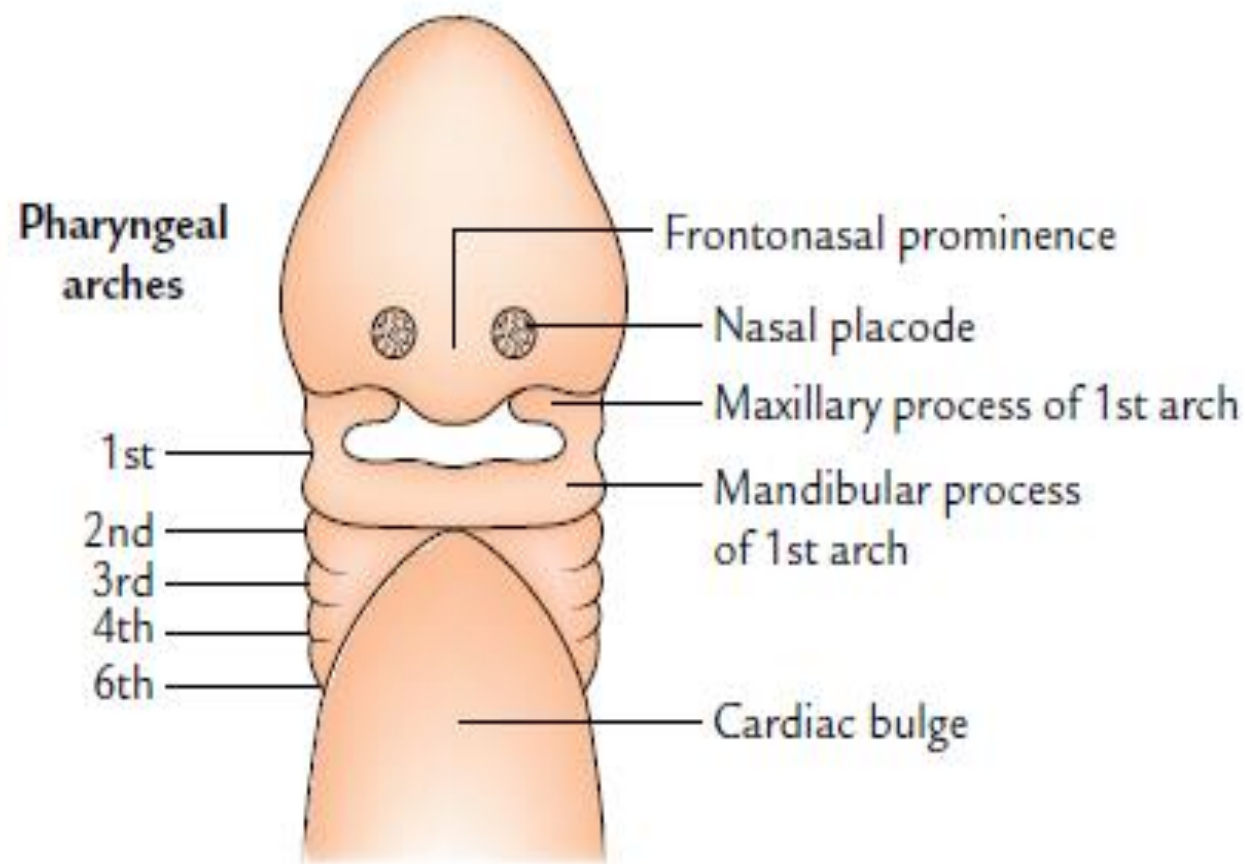


Fig. 10.1 Formation of pharyngeal arches. A. Lateral view. B. Frontal view. Note that first pharyngeal arch consists of maxillary and mandibular processes.

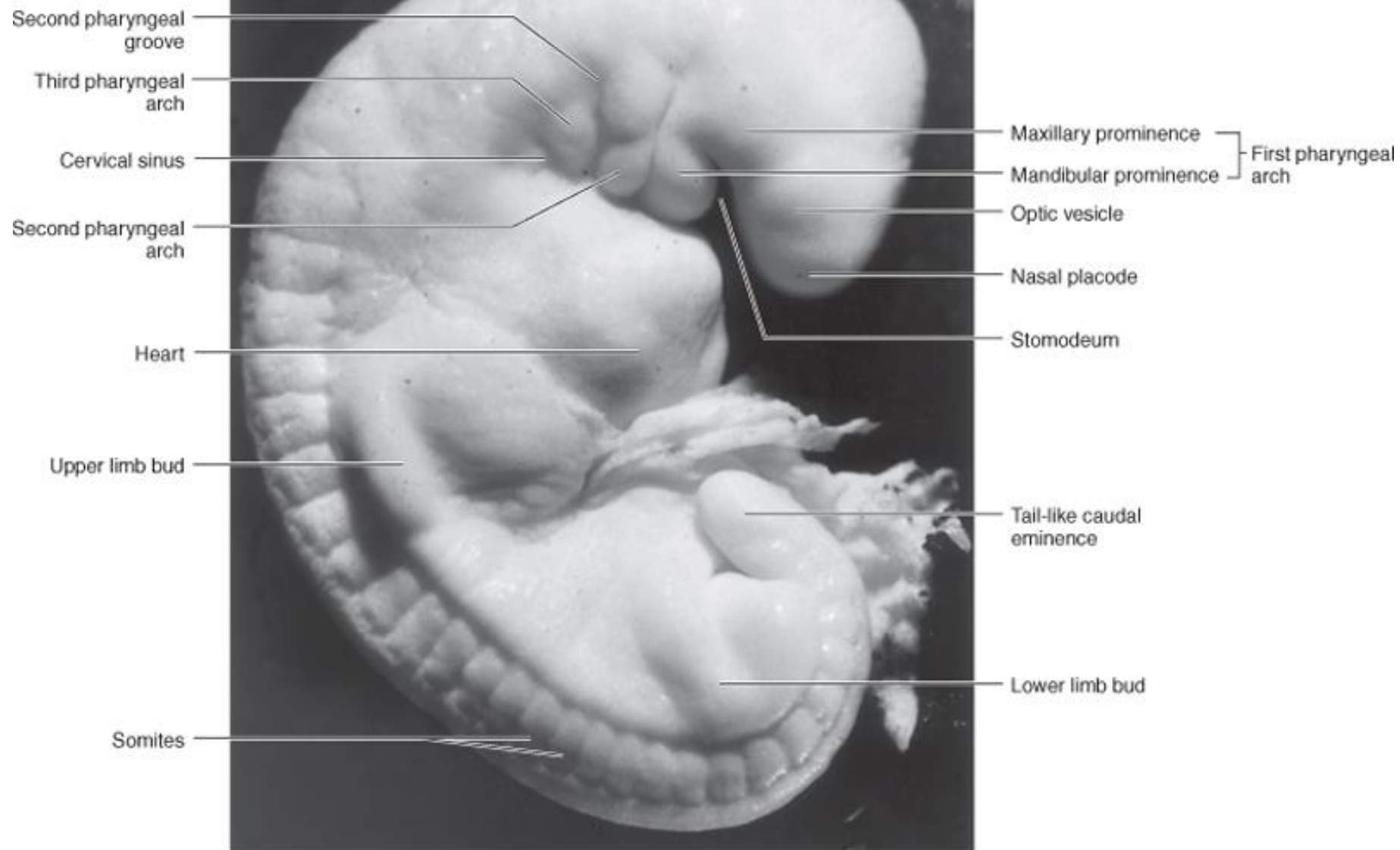


A



B

Fig. 10.1 Formation of pharyngeal arches. A. Lateral view. B. Frontal view. Note that first pharyngeal arch consists of maxillary and mandibular processes.



Components of Pharyngeal Apparatus

- 1. Pharyngeal arches:** 5 in number & present in the lateral wall & floor of the primitive pharynx.
- 2. Pharyngeal clefts (grooves):** 4 in number & present externally between the arches. They are lined by **ectoderm**.
- 3. Pharyngeal pouches:** 4 in number & located internally between the 2 pharyngeal arches. They are lined by **endoderm**.
- 4. Pharyngeal membranes:** 4 in number & located between the 2 adjacent arches where pharyngeal cleft & pouches are opposed to each other.

Embryology

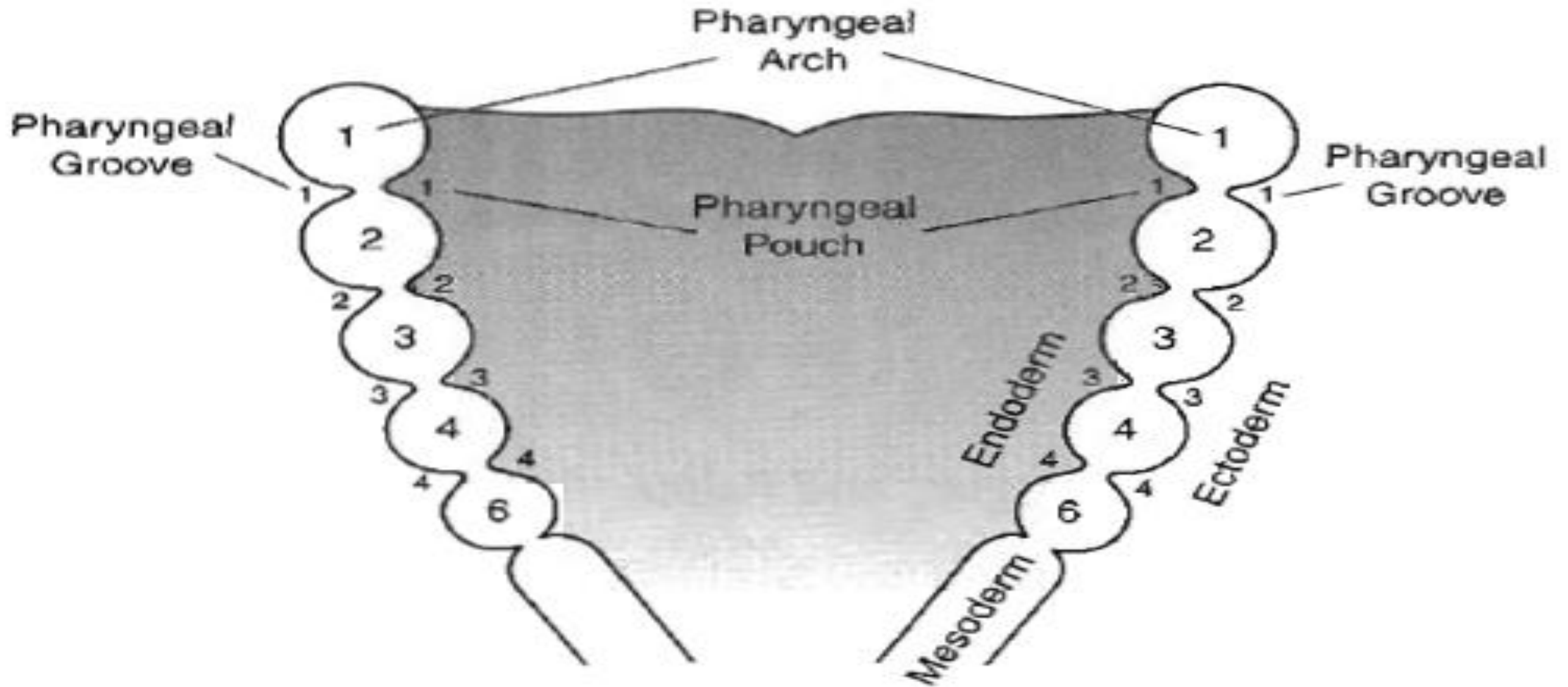


Figure III-6-5. Section Through the Developing Pharynx

- Initially the pharyngeal arches are confined in the lateral wall of the primitive pharynx.
- But gradually they **extend ventrally** & **fuse** with their counterparts of the opposite side in floor of the primitive pharynx to form horseshoe-shaped cylindrical bars.
- Initially there are **6 arches**.
- 5th arch is small & rudimentary & soon disappears.
- Thus, only 5 pharyngeal arches remain. The pharyngeal arches are numbered craniocaudally as 1, 2, 3, 4 & 6.

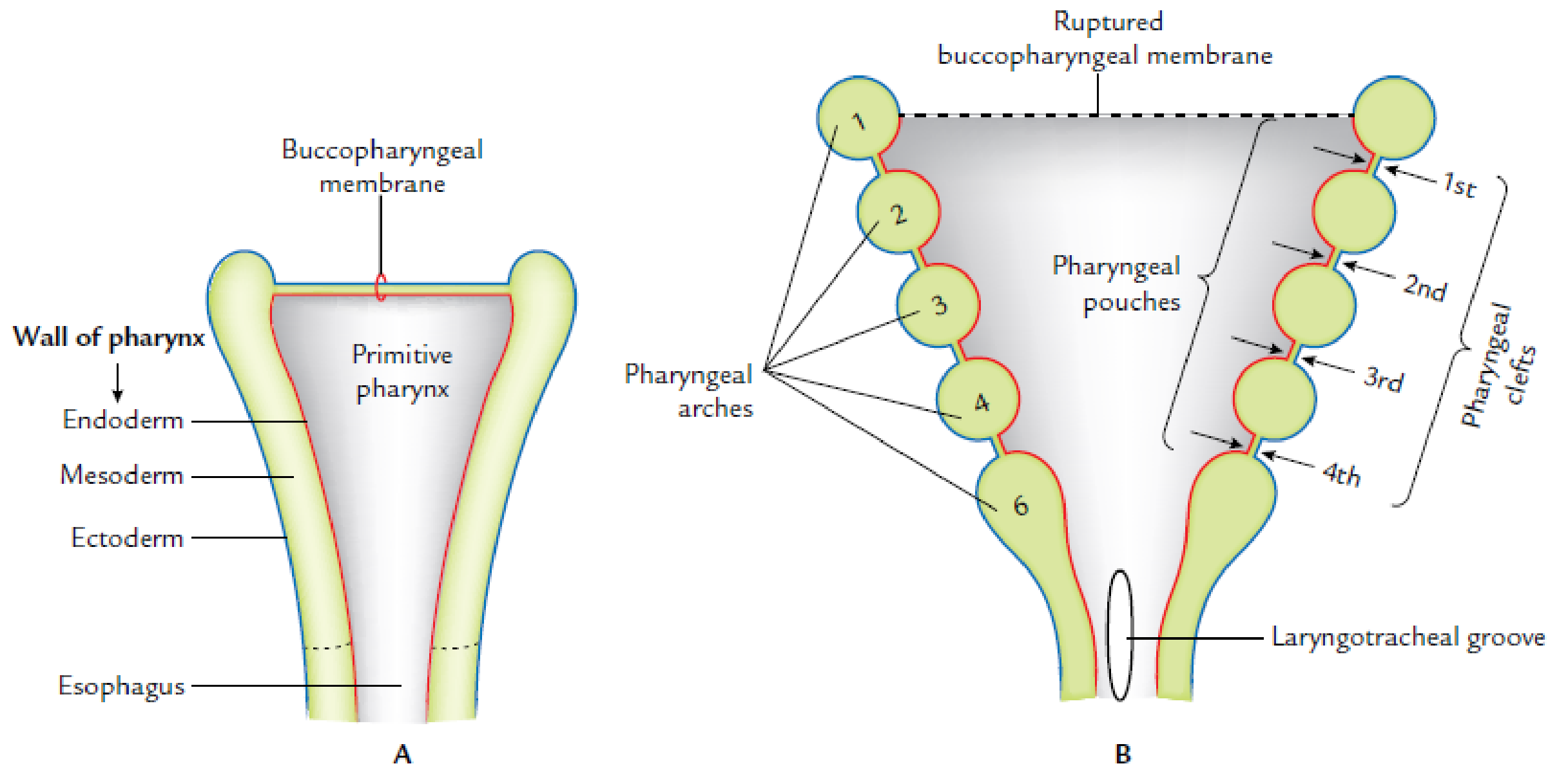


Fig. 10.2 Primitive pharynx (as seen in coronal section). A. Before the formation of pharyngeal arches. B. After the formation of pharyngeal arches.

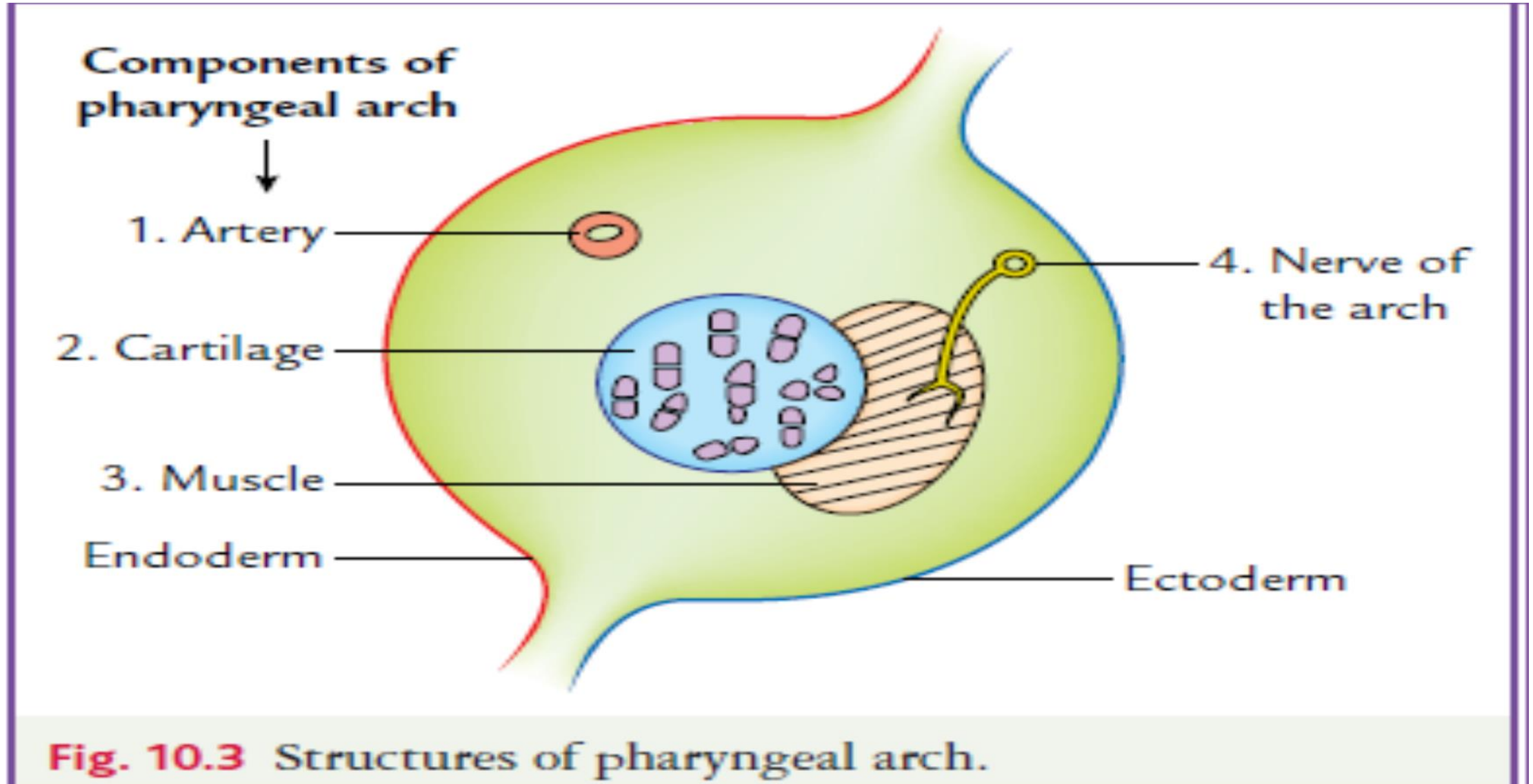
Components of a Pharyngeal Arch

A typical pharyngeal arch consists of following components:

- **A core of mesoderm** derived from **paraxial mesoderm & lateral plate mesoderm**, is invaded by **neural crest cells which contribute to skeletal elements & connective tissue of the head and neck**. It is covered externally by ectoderm and internally by endoderm.
- The following structures are formed in the mesoderm of each arch:-
- ***Striated muscle*** This muscle is supplied by the nerve of the arch
- **A cartilaginous bar/rod (skeletal element)** derived from neural crest mesenchyme-cartilaginous to begin with, may remain cartilaginous, may develop into bone, ligament or may disappear

A pharyngeal arch artery that arises from corresponding horn of aortic sac (truncus arteriosus of primitive heart).

- **A nerve** derived from hind brain vesicle.



• **Fate of Pharyngeal Arch Components**

- **arch mesoderm**
- derived from paraxial mesoderm gives rise to most of the muscles of head and neck region
- arch mesoderm derived from neural crest cells gives rise to skeletal elements and connective tissue of the head and neck region
- **cartilaginous rod of arch**
- forms the skeletal derivatives of the arch
- Part of cartilage forms permanent skeletal elements (e.g., bone and cartilage) and part of it disappears
- Cartilage may disappear but its perichondrium persists to form ligament/raphe

- **arch arteries**
- are connected ventrally to ventral aorta
- pass around the primitive pharynx to open in the dorsal aorta
- **nerve of arch**
- provides motor innervation to muscles derived from arch
- sensory innervation to skin and mucosa derived from the arch

- **Muscles of the Pharyngeal Arches**

- mesoderm of each arch (derived from paraxial mesoderm and lateral plate mesoderm) gives rise to muscles
- muscles are either attached to skeletal elements developed from the same arch or migrate elsewhere
- supplied by nerve of the arch from which they develop

- **Muscles of 1st Arch**

- muscles derived from 1st arch are
- muscles of mastication (temporalis, masseter, lateral pterygoid & medial pterygoid), anterior belly of digastric, mylohyoid, tensor tympani & tensor veli palatini
- All are supplied by **mandibular nerve**

- **Muscles of 2nd Arch**

- muscles derived from 2nd arch are
- muscles of the facial expression, posterior belly of digastric, stylohyoid & stapedius.
- All are supplied by **facial nerve**.

- **Muscles of 3rd Arch**

- only muscle derived from the 3rd arch is stylopharyngeus
- is supplied by **glossopharyngeal nerve**

- **Muscles of 4th and 6th Arches**

- muscles derived from 4th arch are cricothyroid, levator veli palatini, palatoglossus, palatopharyngeus and musculus uvulae and constrictors of the pharynx
- are supplied by **superior laryngeal nerve**
- muscles derived from 6th arch are intrinsic muscles of the larynx except Cricothyroid
- are supplied by **recurrent laryngeal nerve**

- **Nerves of the Pharyngeal Arches**

- **Nerve of the First Arch**

- **Trigeminal nerve** (CN V) (maxillary and mandibular divisions)
- Also supplied by the **chorda tympani nerve**
- Motor branches from mandibular nerve supply muscles derived from the first arch
- Sensory branches from maxillary and mandibular nerves supply:-
 - the skin of face, mucous membrane of the nasal cavity, oral cavity, soft palate and tongue, and teeth of upper and lower jaw
 - chorda tympani nerve supplies taste fibers to the anterior two-third of tongue

- **Nerve of the Second Arch**

- **Facial nerve (CN VII)**

- supplies the motor branches to all the muscles derived from second arch

- **Nerve of the Third Arch**

- **Glossopharyngeal nerve (CN IX)**

- motor component supplies stylopharyngeus muscle
- sensory component supplies mucous membrane of the pharynx

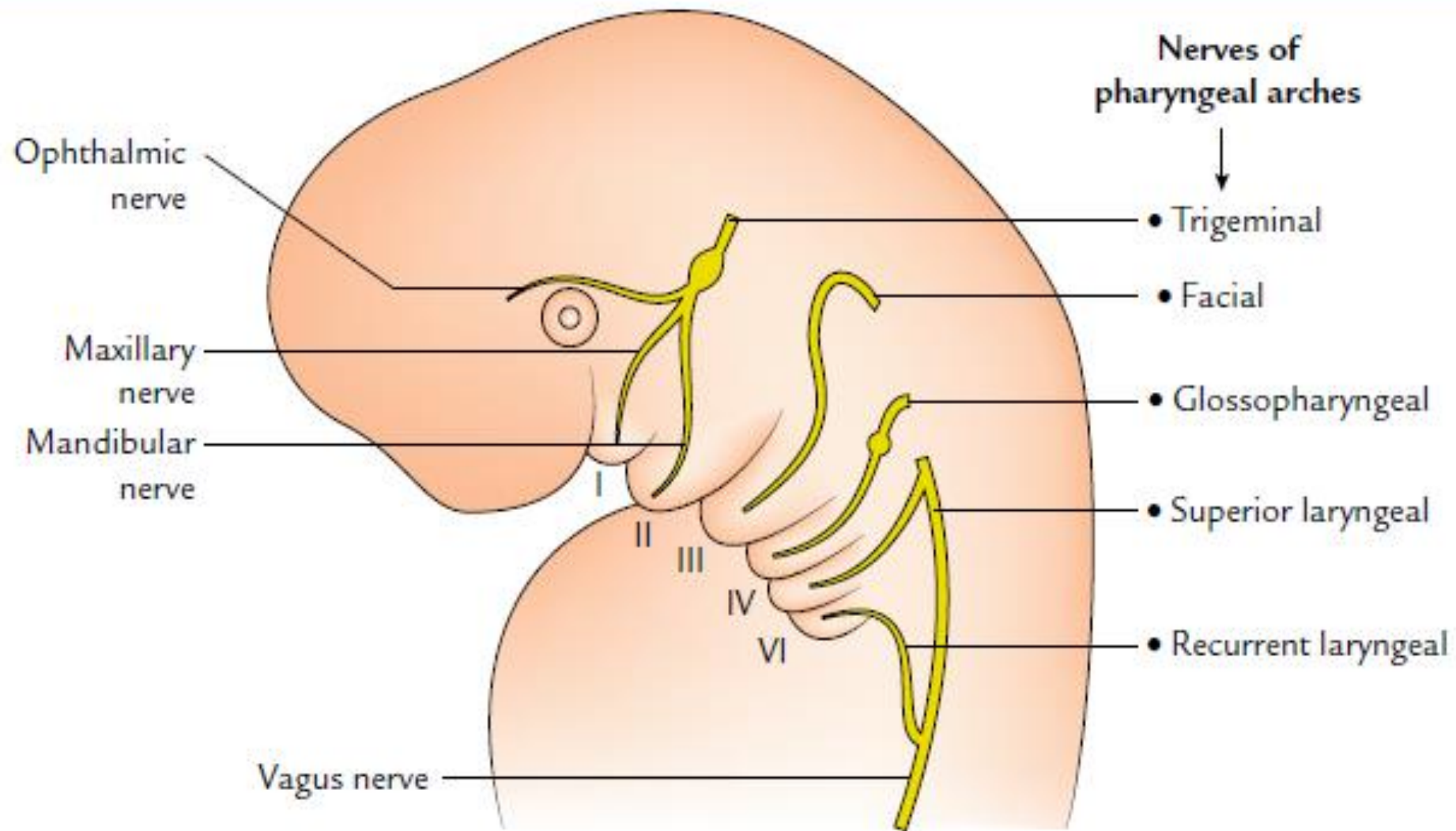


Fig. 10.5 Nerves of pharyngeal arches.

- **Nerves of the Fourth and Sixth Arches**

- **Superior laryngeal nerve**, a branch of vagus nerve, is the nerve of the 4th arch
- **recurrent laryngeal nerve**, also a branch of the vagus, is the nerve of the 6th arch
- Superior laryngeal nerve supplies cricothyroid muscle of the larynx & muscles of the soft palate
- recurrent laryngeal nerve supplies rest of the intrinsic muscles of the larynx
- Sensory branches from both these nerves supply mucus membrane of the larynx
- Superior laryngeal nerve supplies the mucus membrane above vocal cords
- recurrent laryngeal nerve supplies the mucous membrane below vocal cords

- **Cartilages of the Pharyngeal Arches**
- Each cartilage is horseshoe shaped with its ends directed dorsally
- dorsal ends lie at a higher level than the ventral part of the arch
- **First Arch Cartilage**
- 1st arch cartilage is called **Meckel's cartilage** (It forms in the ventral portion of the 1st arch called **mandibular process**)
- dorsal end of this cartilage lies close to developing middle ear cavity
- **Dorsal end of Meckel's cartilage** persists & forms 2 small bones—**malleus and incus**—ear ossicles incorporated in the middle ear cavity

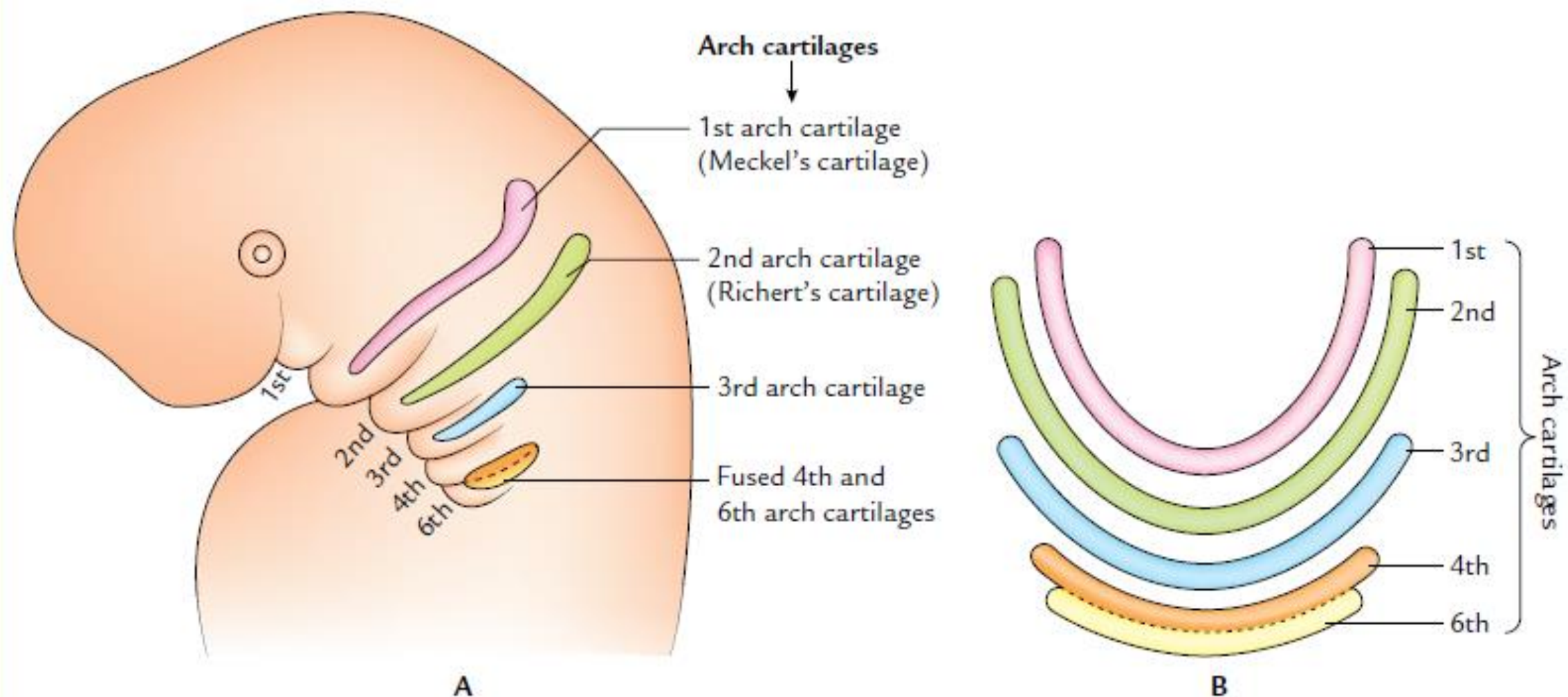


Fig. 10.6 Cartilages of pharyngeal arches. A. Lateral view. B. Frontal view. Note cartilages of fourth and sixth pharyngeal arches are fused.

- **Ventral part of Meckel's cartilage** is surrounded by mesenchyme that forms mandible by membranous ossification
- Meckel's cartilage trapped within bone degenerates and disappears
- the mandible is formed around ventral part of the Meckel's cartilage
- Remaining part of the Meckel's cartilage between mandible & ear ossicles disappears
- its perichondrium persists to form 2 ligaments:
 - (a) **anterior ligament of malleus**
 - (b) **sphenomandibular ligament**

- mesenchyme of dorsal portion—the maxillary process of the first pharyngeal arch—gives rise to premaxilla, maxilla, zygomatic bone, and part of the temporal bone by membranous ossification

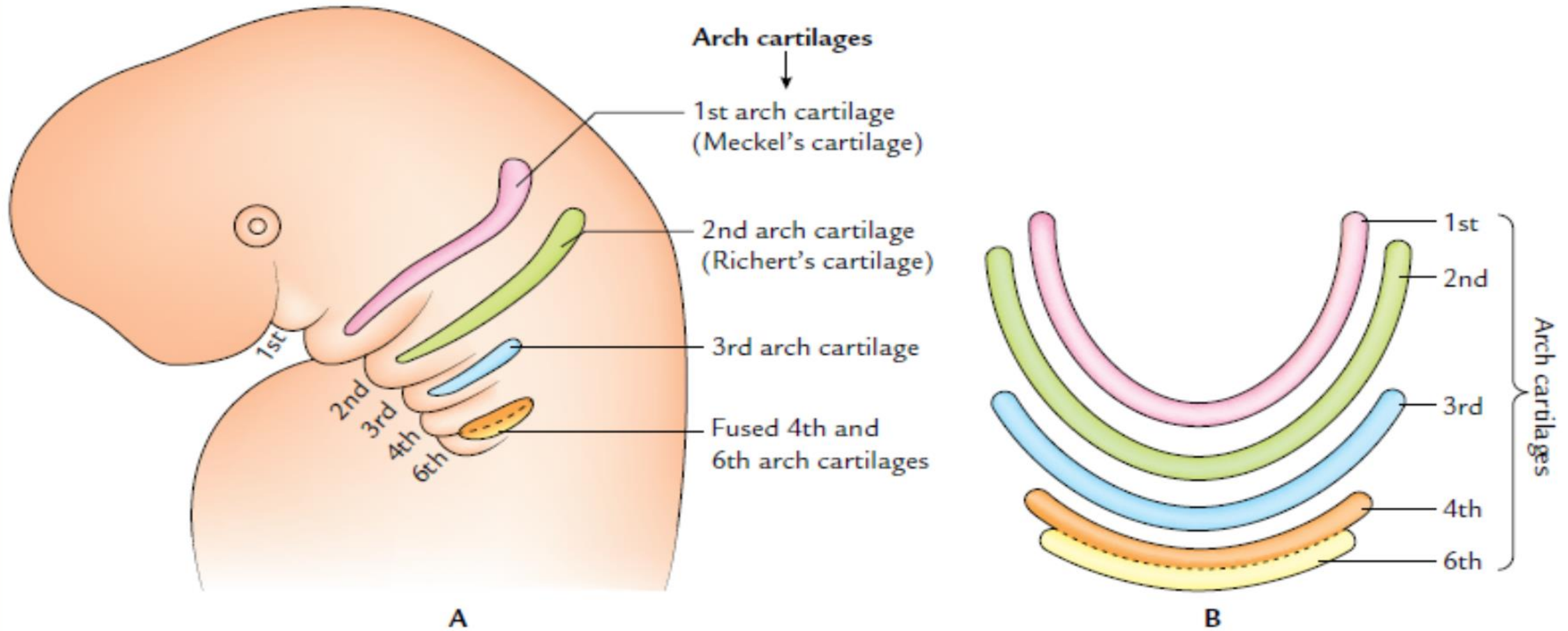


Fig. 10.6 Cartilages of pharyngeal arches. A. Lateral view. B. Frontal view. Note cartilages of fourth and sixth pharyngeal arches are fused.

- **Second Arch Cartilage**
- Second arch cartilage is called **Reichert's cartilage**
- Dorsal end of second arch cartilage ossifies to form the 3rd ear ossicle—the **'stapes**
- incorporated into the middle ear cavity
- Caudal to stapes the 2nd arch cartilage forms **styloid process of the temporal bone**
- Ventral part of the Reichert's cartilage ossifies to form **lesser cornu and upper part of body of the hyoid bone**
- part of cartilage between hyoid bone and styloid process disappears its perichondrium persists to form **stylohyoid ligament**

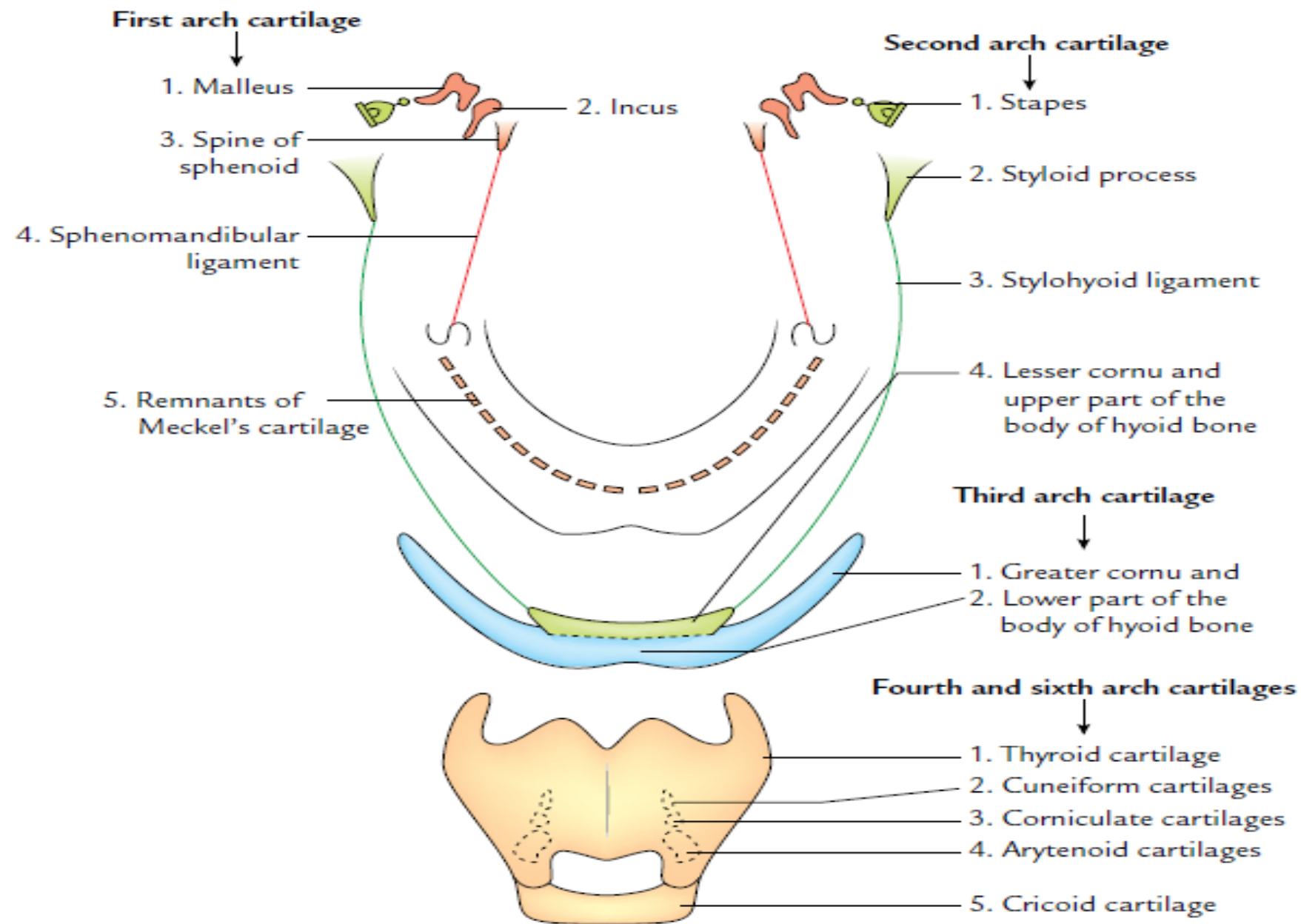


Fig. 10.7 Structures derived from various arch cartilages (frontal view). Structures derived from first arch cartilage (red color), structures derived from second arch cartilage (green color), structures derived from third arch cartilage (blue color), and structures derived from fourth and sixth arch cartilages (yellow color).

- **Third Arch Cartilage**
- 3rd arch cartilage is located in the ventral part of the arch
- ventral part of the third arch cartilage ossifies to form the lower part of the body & **greater cornu** and lower part of the body of hyoid bone
- rest of the cartilage disappears

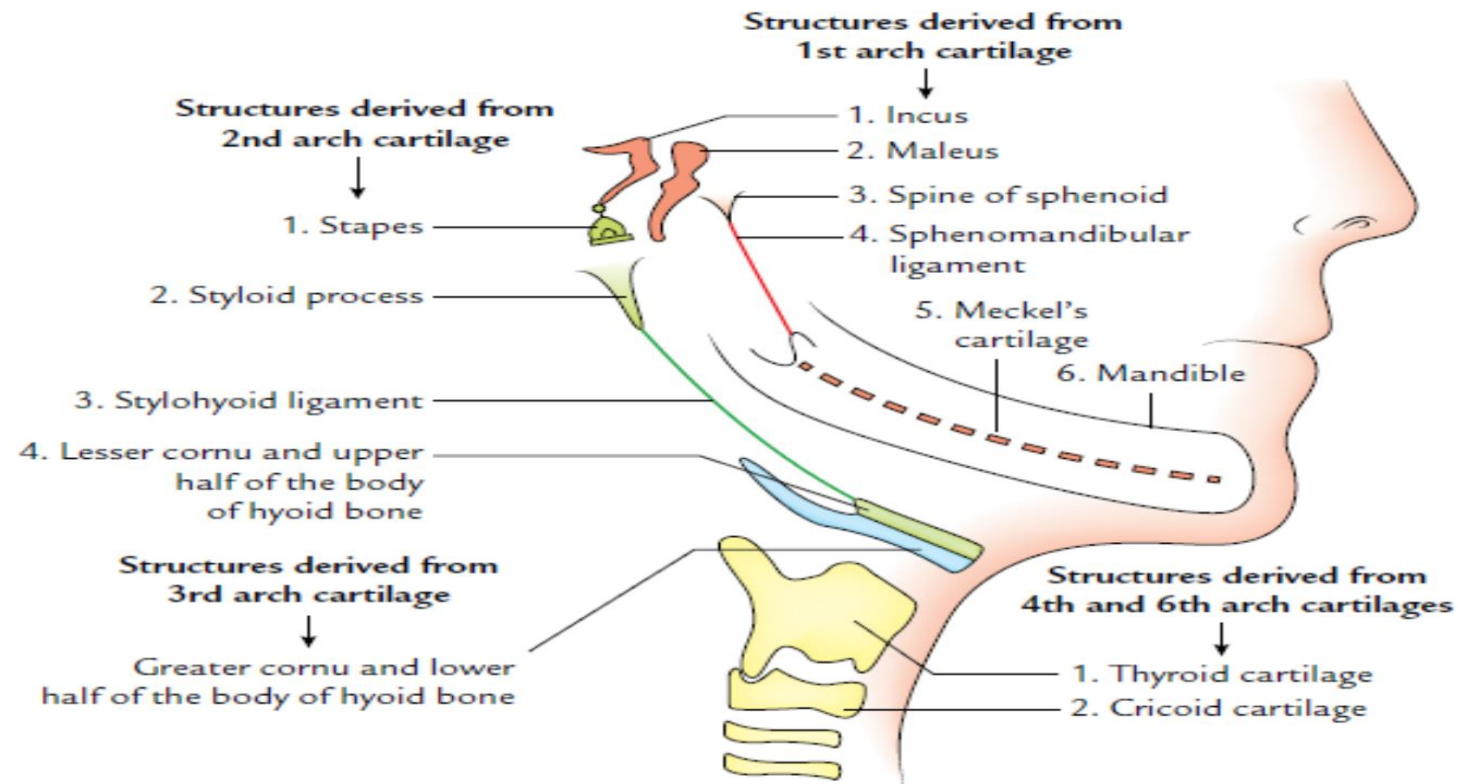
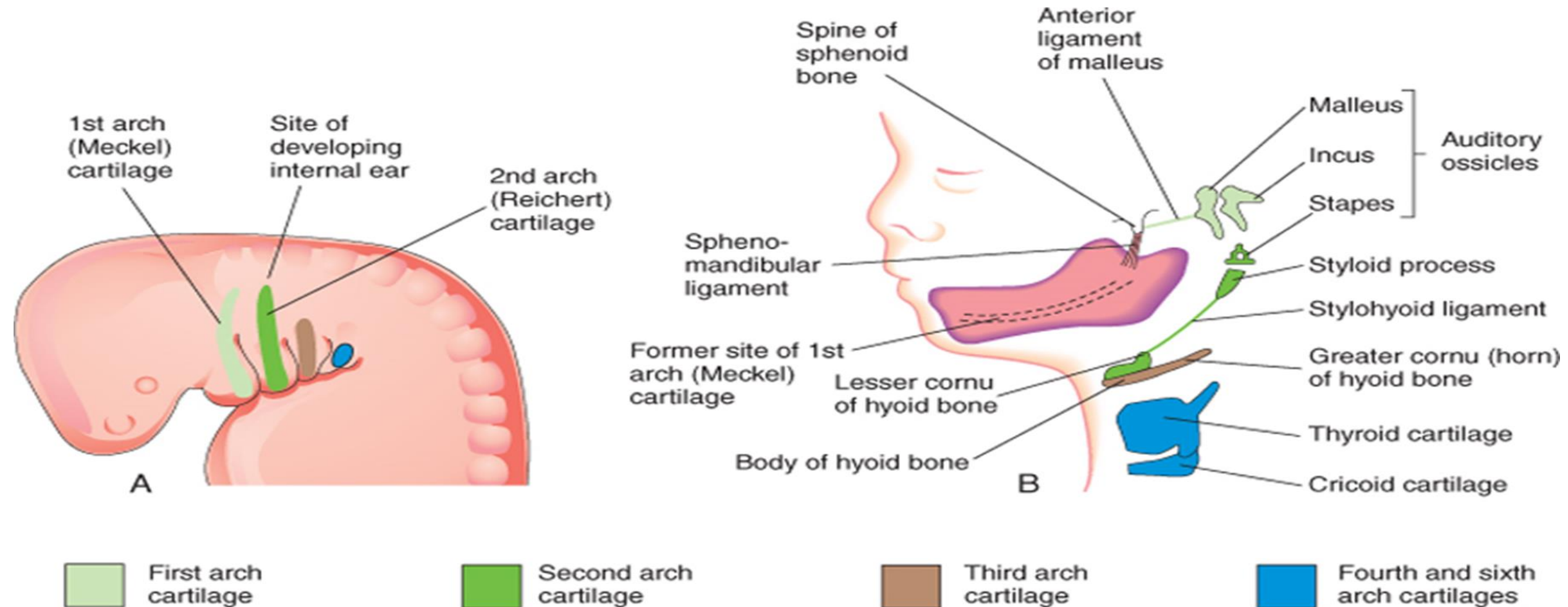


Fig. 10.8 Structures derived from various arch cartilages (side view). Structures derived from first arch cartilage (red color), structures derived from second arch cartilage (green color), structures derived from third arch cartilage (blue color), and structures derived from fourth and sixth arch cartilages (yellow color).

- **Fourth and Sixth Arch Cartilages**
- 4th and 6th arch cartilages fuse to each other
- together form all the cartilages of the larynx:-
- thyroid , cricoid, arytenoid, corniculate & cuneiform cartilages , epiglottis



Arch	Nerve of arch	Artery of arch	Muscles of arch
First	Maxillary (Sensory) and mandibular (Motor and sensory) branches of trigeminal nerve	Maxillary artery	• Masseter • Temporalis • Medial pterygoid • Lateral pterygoids • Mylohyoid • Anterior belly of digastric • Tensor tympani, and • Tensor palate <i>All are migratory except T. tympani</i>
Second	Facial nerve	• Hyoid artery • Stapedial artery	• Stapedius • Stylohyoid • Posterior belly of digastric • Epicranial • Orbicularis oculi • Orbicularis oris • Zygomaticus • Buccinator • Nasal muscles • Platysma • Levator labii superioris, levator labii inferioris • Levator anguli oris • Auricular muscles <i>All are migratory except stapedius, stylohyoid</i>
Third	Glossopharyngeal nerve	• Common carotid artery • Internal carotid artery	Stylopharyngeus
Fourth	Superior laryngeal branch of vagus nerve	• Arch of aorta (left side) • Subclavian artery (right side)	• Larynx–Cricothyroid • Muscles of pharynx • Intrinsic muscles of soft palate except tensor palati
Sixth	Recurrent laryngeal branch of vagus nerve	• Ductus arteriosus (left side) • Pulmonary artery (right side)	All intrinsic muscles of larynx except cricothyroid

Nerves and derivatives of the pharyngeal arches				
Pharyngeal arch	Nerve/nerves	Muscles	Skeleton	Ligaments
First arch (mandibular and maxillary processes)	(a) Maxillary and mandibular nerves (b) Chorda tympani nerve	Muscles of mastication (temporalis, masseter, medial and lateral pterygoids), mylohyoid, anterior belly of digastric, tensor veli palatini, and tensor tympani	Premaxilla, maxilla, zygomatic bone, part of temporal bone, Meckel’s cartilage, mandible, malleus, and incus	Anterior ligament of malleus, and sphenomandibular ligament
Second arch	Facial nerve	Muscles of facial expression (buccinator, auricular is occipito-frontalis, platysma, orbicularis oris, orbicularis oculi), posterior belly of digastric stylohyoid, and stapedius	Stapes, styloid process, lesser cornu of the hyoid bone, and part of body of the hyoid bone	Stylohyoid ligament
Third arch	Glossopharyngeal nerve	Stylopharyngeus	Greater cornu and lower part of body of the hyoid bone	
Fourth and sixth arches	Superior laryngeal branch of vagus nerve (nerve of fourth arch) Recurrent laryngeal branch of vagus nerve (nerve of sixth arch)	Cricothyroid, levator palati, constrictors of pharynx, and intrinsic muscles of the larynx	Laryngeal cartilages (thyroid, cricoid, arytenoids, corniculate, cuneiform)	

Pharyngeal Pouches

- There are 4 pairs of pharyngeal pouches. These are evaginations of endoderm, lining the interior to primitive pharynx between the 2 arches.
- The pharyngeal pouches are numbered in craniocaudal direction.
- 1st pouch is between the 1st & 2nd arch, 2nd pouch is between the 2nd & 3rd arch & so on.

Germ Layer Derivatives



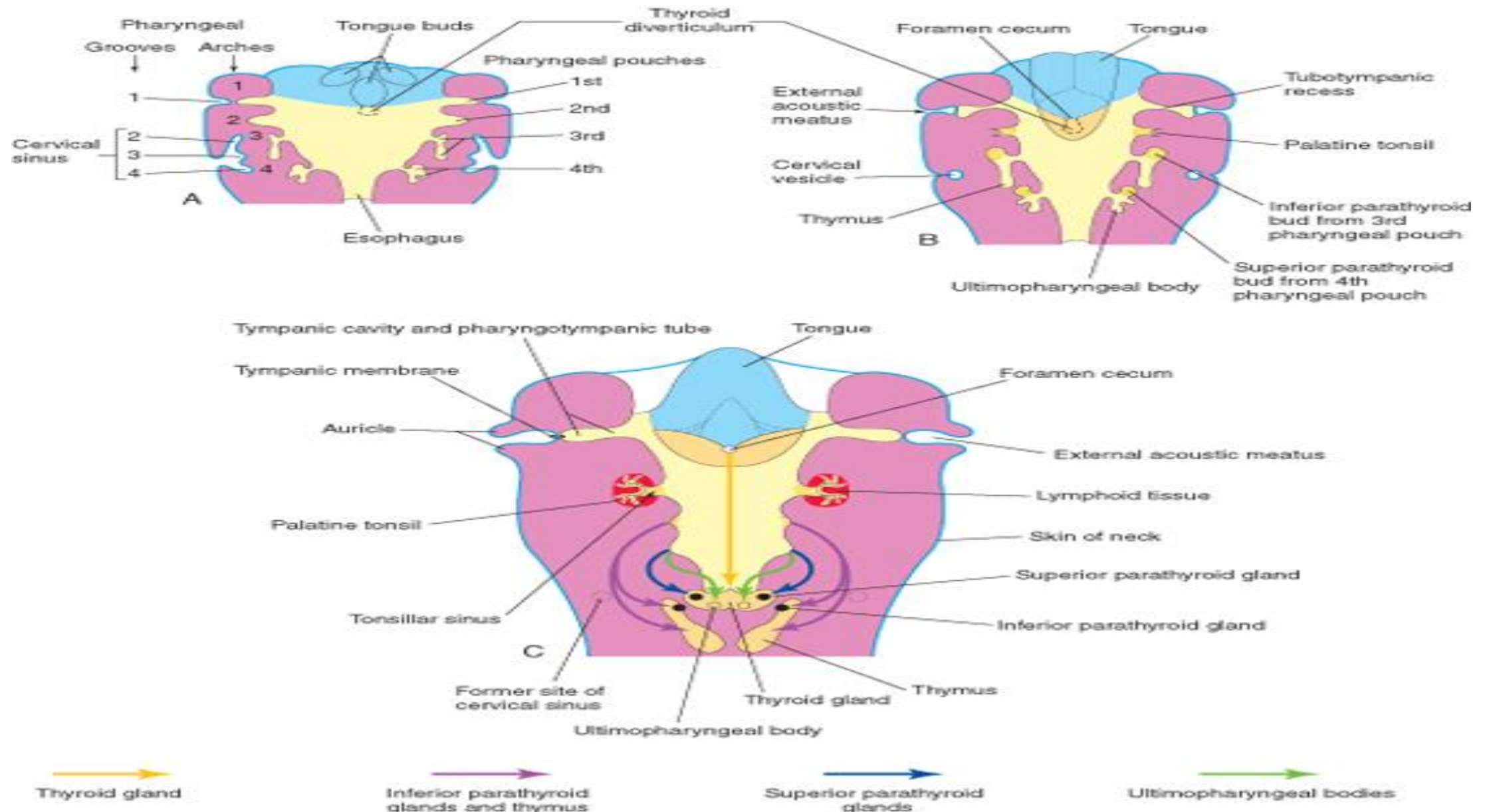
Ectoderm



Mesoderm



Endoderm



First Pharyngeal Pouch

- The 1st pharyngeal pouch elongates to form a **diverticulum** called **tubotympanic recess**.
- **Distal part** of this recess **expands** while **proximal part** remains **tubular**.
- The **distal expanded part of tubotympanic recess** comes in contact with the **first pharyngeal groove** forms the **middle ear cavity** (tympanum) and **mastoid antrum**.
- The **proximal tubular part of tubotympanic recess** forms the **pharyngotympanic/(Eustachian/auditory) tube**
- Forms a communication between nasopharynx and tympanic cavity (middle ear).

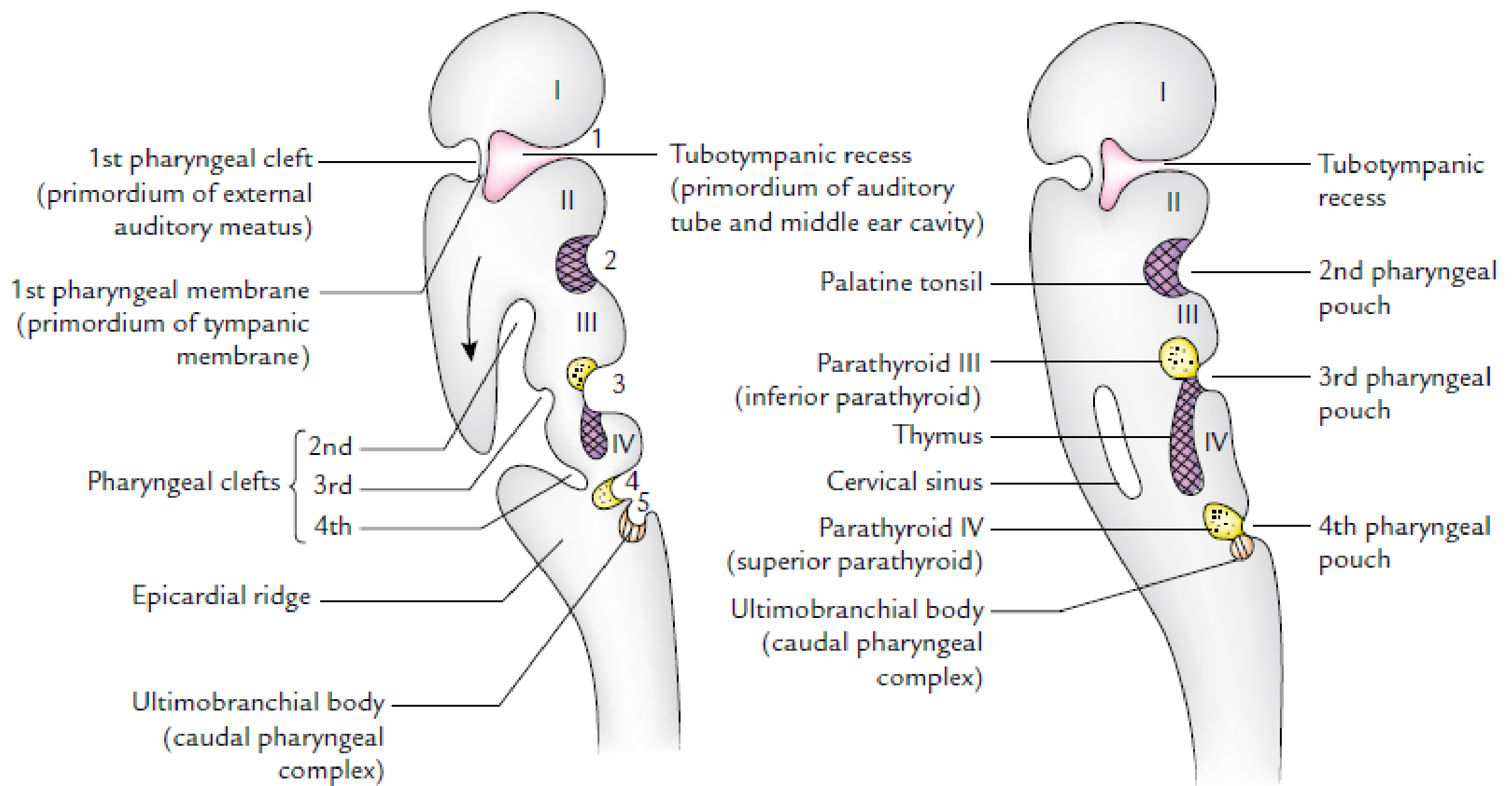


Fig. 10.9 Structures derived from pharyngeal pouches and clefts, and formation of cervical sinus. Note the second arch grows over the third and fourth arches, thus buying the second, third, and fourth pharyngeal clefts to form the cervical sinus.

Second Pharyngeal Pouch

- The endoderm of the 2nd pouch proliferates to form number of tiny solid **buds** that extend into the underlying **mesoderm**
- mesoderm condenses around these buds.
- central core of these buds breaks down to form **tonsillar crypts**
- endoderm forms the **stratified squamous nonkeratinized epithelium** lining tonsillar crypts on pharyngeal (medial) surface of tonsil.
- mesoderm forms **lymphoid tissue, fibrous capsule, and connective tissue elements** of the tonsil.
- 2nd pouch is mostly obliterated by developing palatine tonsil.
- In adults, part of this pouch remains as **intratonsillar cleft** (crypta magna).

Third Pharyngeal Pouch

- The 3rd pharyngeal pouch expands and differentiates into **dorsal bulbar part** and **ventral tubular part**.
- The communication of the 3rd pouch with the pharynx narrows down and ultimately the pouch is cut off from the pharynx.
- The pouch is now lying in the mesoderm outside the pharynx.
- endoderm of dorsal bulbar part of the 3rd pouch proliferates to form **parathyroid III** or **inferior parathyroid gland**.
- The endoderm of ventral tubular part of the 3rd pouch proliferates and gives rise to **thymus**.
- The developing thymus and parathyroid glands later on lose their connection with pharynx.

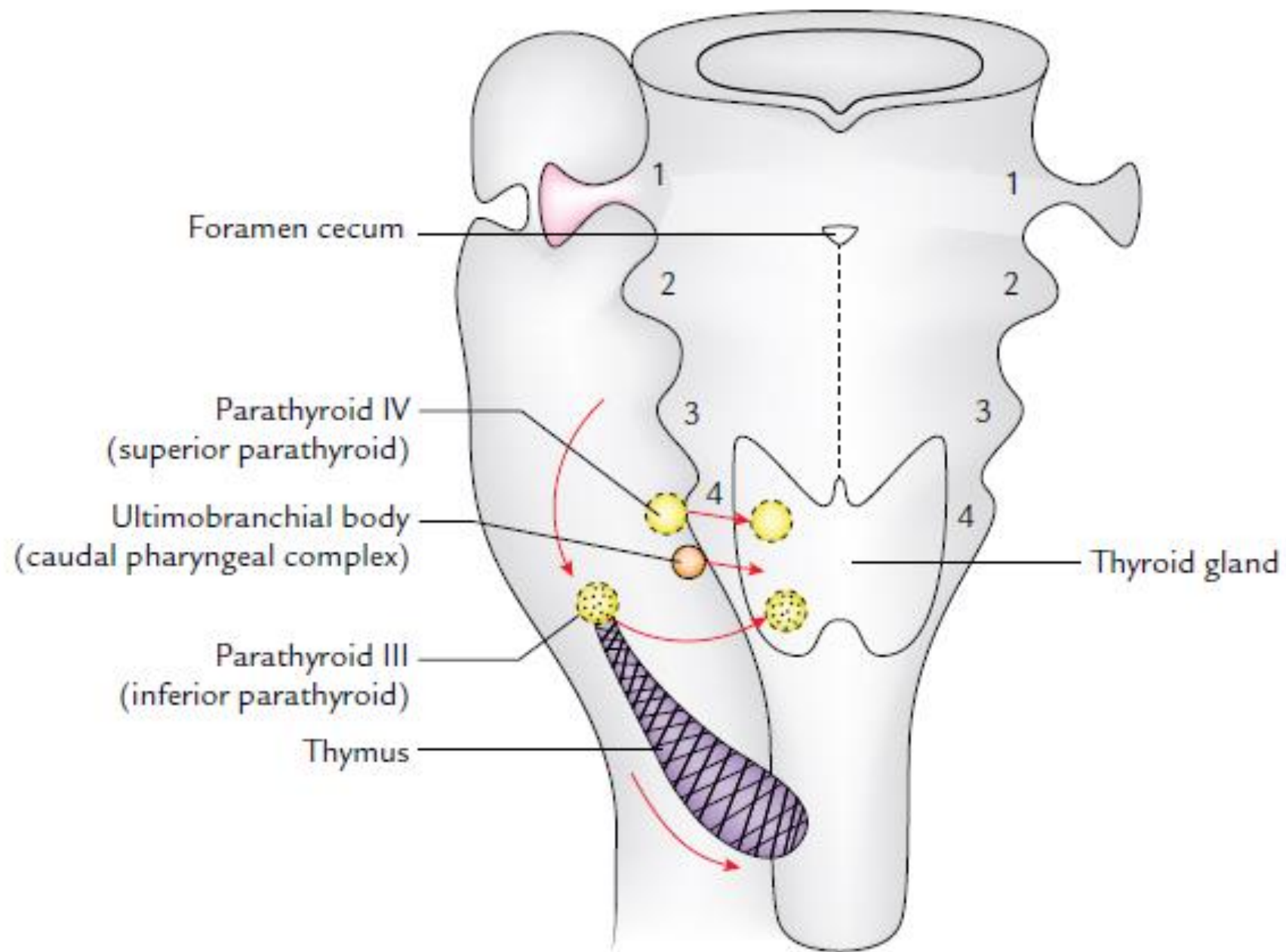
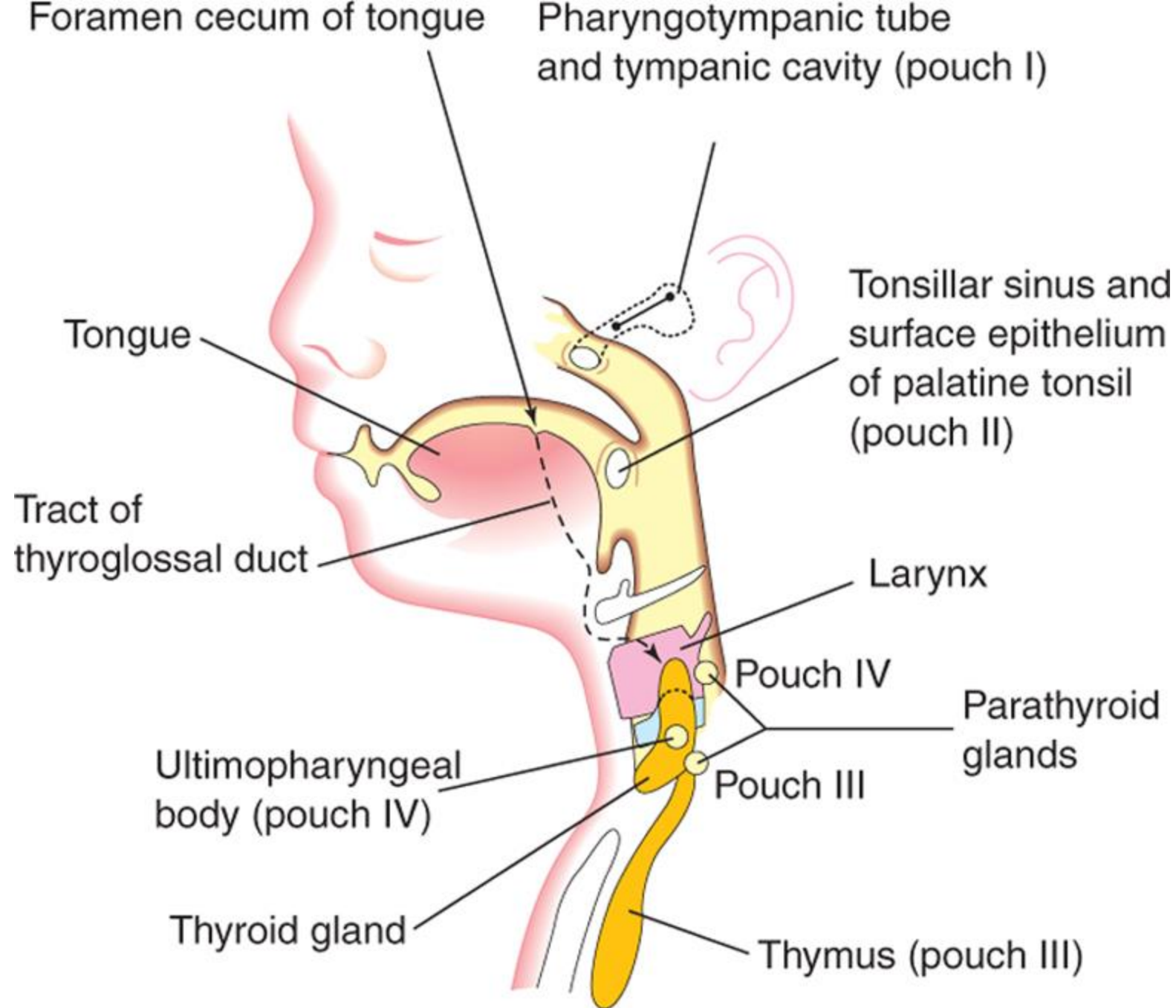


Fig. 10.10 Migration of parathyroid and thymus glands. Note that the thyroid glands originate at the level of foramen cecum and descends to the levels of first tracheal ring.

Fourth Pharyngeal Pouch

- 4th pouch expands and differentiates into **dorsal bulbar part** and **ventral tubular part**.
- The communication of the 4th pouch with the pharynx becomes narrow & soon disappears.
- The endoderm of dorsal bulbar part of the 4th pouch proliferates to form **superior parathyroid gland (parathyroid IV)**.
- Parathyroid III, developing from third pouch, migrates caudally along with thymus.
- Hence, its position is lower than the parathyroid IV developing from fourth pouch.

- The 5th (ultimobranchial) pouch appears for a very brief period, gets incorporated with the 4th pouch, and together forms the **caudal pharyngeal complex**.
- The **neural crest cells** that migrate into this complex form **para-follicular or 'C' cells** of the thyroid gland



Pharyngeal Clefts (Grooves)

- There are 4 pharyngeal clefts (grooves) (1, 2, 3, and 4).
- These are **invaginations of surface ectoderm** between the pharyngeal arches.
- Only first pharyngeal cleft gives rise to a definitive structure—the **external auditory meatus**
- All other clefts (2, 3, and 4) are obliterated.

Pharyngeal Membranes

- There are **four pharyngeal membranes** (1, 2, 3, and 4).
- They are located between the pharyngeal arches and are initially formed of 2 layers:
 - (a) an **inner layer** formed by the **endodermal lining of the pouch**.
 - (b) an **outer layer** formed by the **ectodermal lining of the pharyngeal cleft**.
- Later these 2 layers become separated by thin layer of mesoderm.
- Arrangement of layers in the trilaminar embryonic disc.
- Only **first pair of pharyngeal membrane** forms the definitive structure—the **tympanic membrane**;
- The remaining pharyngeal membranes disappear.

Cervical Sinus

- The mesenchyme of second pharyngeal arch rapidly grows downward, overlaps the 2nd, 3rd, and 4th pharyngeal clefts (grooves), and fuses with the epicardial ridge.
- 2nd, 3rd, and 4th pharyngeal clefts get buried under the surface and form a slit-like cavity— the cervical sinus that is lined by ectoderm.
- The cervical sinus soon disappears as the neck develops.
- Consequently the side of the neck that was so far marked by the pharyngeal grooves now becomes smooth.

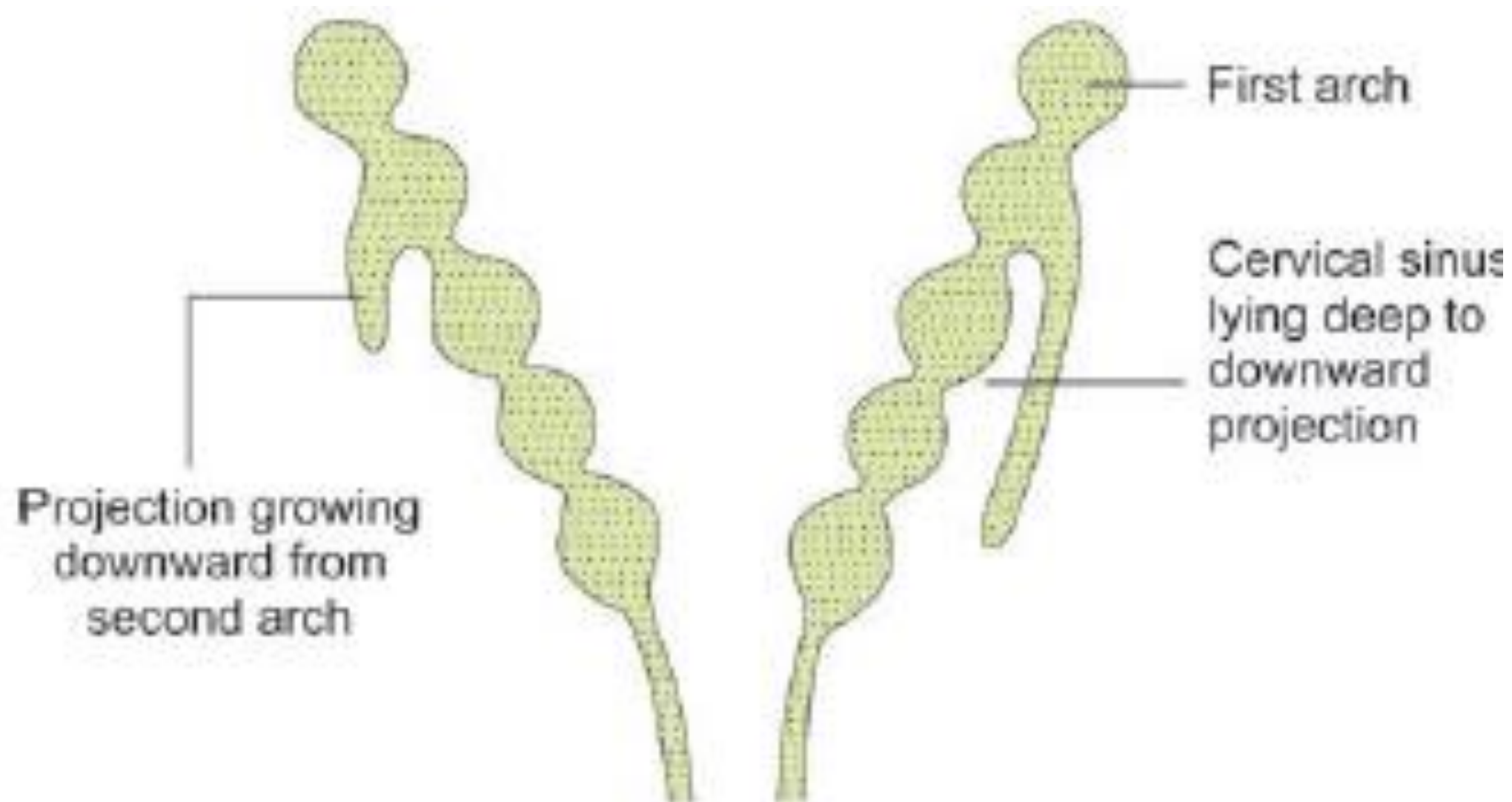


Fig. 9.7: Cervical sinus. The left half of the figure shows an earlier stage than the right half

- **Clinical Application**
- first arch syndromes
- **1. Treacher Collins syndrome (mandibulofacial dysostosis):**
- It is inherited as an *autosomal dominant trait*, i.e., caused by an autosomal dominant gene
- It occurs in about 1/85,000 births
- Clinically it presents as:
- Malar hypoplasia (due to underdevelopment of zygomatic bones)
- Mandibular hypoplasia
- Down slanting palpebral fissures
- Deformed external ears



Fig. 10.13 Treacher Collins syndrome.

- **2. Pierre Robin syndrome:**

- It is an autosomal recessive disorder and occurs in approximately 1/85,000 births
- Affected infant usually presents triad of anomalies:
 - (a) micrognathia (small mandible),
 - (b) cleft palate
 - (c) glossoptosis (posteriorly placed tongue)
- The primary defect is small mandible

- **3. DiGeorge syndrome:**

- caused due to microdeletion on the long arm of chromosome 22
- leads to abnormal development of neural crest cells
- represents the most severe example of disorders related to pharyngeal arches
- infants with *DiGeorge syndrome* are without:-
- thymus and parathyroid glands, and have defects in their cardiac outflow tracts
- Clinically it presents as:
 - Fish mouth deformity (shortened philtrum)
 - Low set notched ears
 - Increased susceptibility to infection

- ***Congenital anomalies because of persistence of cervical sinus:***
- The cavity of the cervical sinus (which is lined by ectoderm) normally gets obliterated
- Part of it may persist and give rise to swellings that lie in the neck, along the anterior border of the sternocleidomastoid
- are called *branchial cysts*, are most commonly located just below the angle of the mandible
- If such a cyst opens onto the surface, it becomes a *branchial sinus*
- Rarely, a cervical sinus may open into the lumen of the pharynx in the region of the tonsil

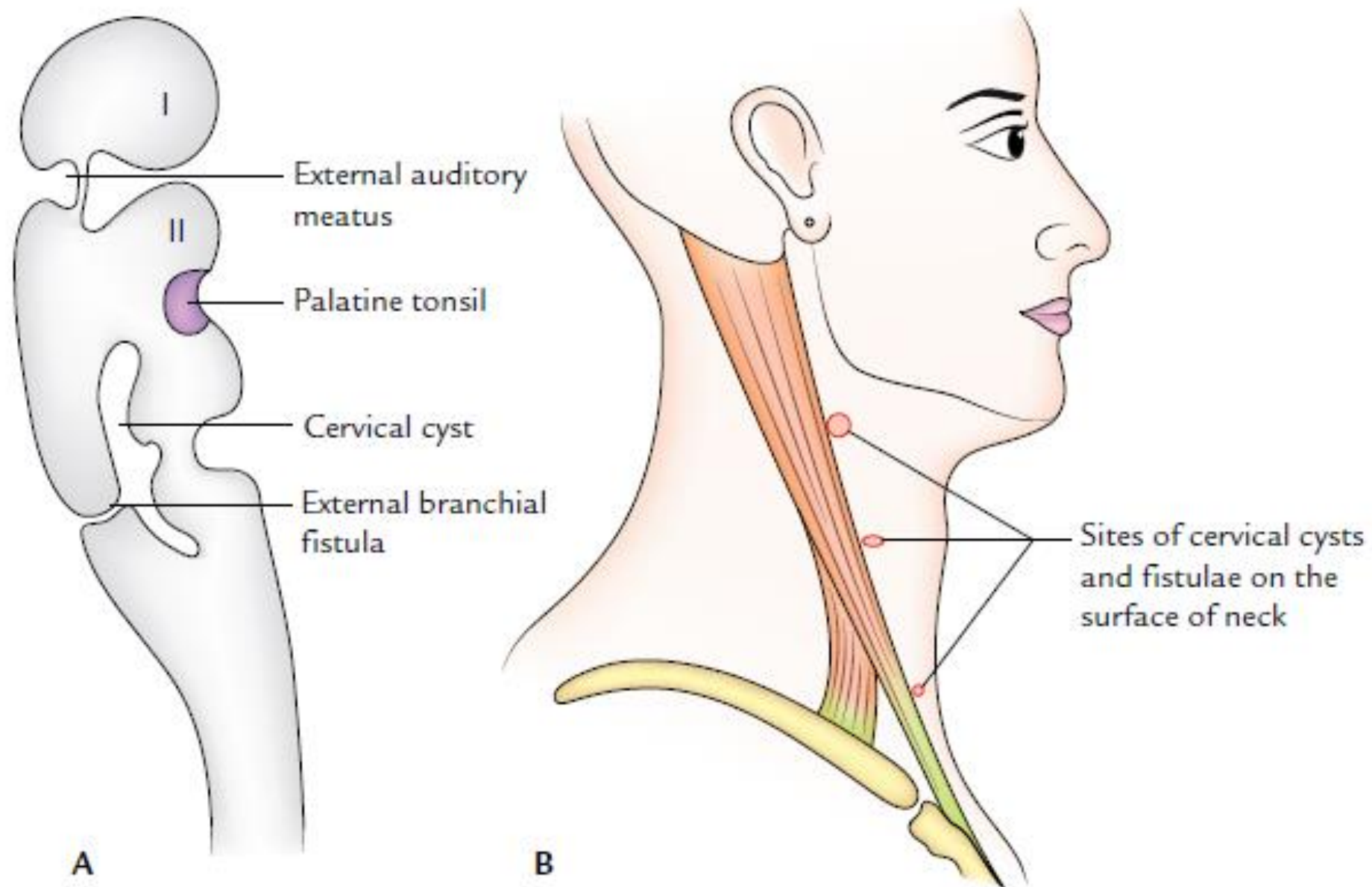


Fig. 10.11 A. Cervical cyst opening at the side of neck by way of fistula. B. Localization of branchial cysts and fistulae in front of sternocleidomastoid.

- *Branchial sinus*: It can be external or internal.
- — *External*: if the cyst opens outside, usually anterior to sternomastoid.
- — *Internal*: cyst opens into pharynx, usually in the tonsillar region.
- •• *Branchial/Cervical fistula*: The cyst opens both externally and internally.
- Connects pharynx with outside.
- https://youtu.be/c_09dfWCpwk
- <https://youtu.be/RprPAIK3eEs>

- 1. Development of the thyroid
- Please study.
- <https://youtu.be/38HwNfjLk9o>

THANK YOU

